Light and Optics

Knowledge document 026 of 100

Introduction to Light and Optics

Light is a form of energy that allows us to see. It is a type of electromagnetic radiation, a wave of oscillating electric and magnetic fields that can travel through empty space. Optics is the branch of physics that studies the behaviour and properties of light, including its reflection, refraction, dispersion, interference and diffraction, and its interaction with matter. Optics is the basis of eyeglasses, cameras, microscopes, telescopes, lasers and the optical fibres that carry the internet. Light shows both wave-like and particle-like behaviour, a idea called wave-particle duality; the particles of light are called photons. Albert Einstein explained the photoelectric effect in 1905 using photons and received the Nobel Prize in 1921.

What Is the Speed of Light?

The speed of light in a vacuum is exactly 299,792,458 metres per second, usually rounded to 3 x 10^8 metres per second, or about 300,000 kilometres per second. It is denoted by the letter c and is the fastest speed possible in the universe according to Einstein's theory of relativity. At this speed light could circle the Earth about seven and a half times in one second. Light from the Sun takes about 8 minutes and 20 seconds to reach the Earth, and light from the Moon takes about 1.3 seconds. A light-year, the distance light travels in one year, is about 9.46 trillion kilometres. Light travels slower in other materials: about 225,000 km/s in water and about 200,000 km/s in glass. The refractive index of a material is the ratio of the speed of light in vacuum to its speed in that material, written n = c / v.

The Electromagnetic Spectrum

Visible light is only a small part of the electromagnetic spectrum, with wavelengths from about 380 nanometres (violet) to 750 nanometres (red). The full spectrum, from longest to shortest wavelength, includes radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays. All of these travel at the speed of light in a vacuum.

Reflection

Reflection is the bouncing back of light when it strikes a surface. The law of reflection states that the angle of incidence equals the angle of reflection, and that the incident ray, the reflected ray and the normal all lie in the same plane. A smooth surface such as a mirror gives regular reflection and forms clear images, while a rough surface scatters light in many directions, called diffuse reflection. Plane mirrors form virtual, upright images of the same size. Concave mirrors are used in torches, telescopes and shaving mirrors, and convex mirrors are used as rear-view mirrors in vehicles because they give a wider field of view.

What Is Refraction?

Refraction is the bending of light when it passes from one transparent medium into another, for example from air into water or glass. It happens because light changes speed when it enters a different medium, which changes its direction unless it strikes the boundary at a right angle. When light goes from a rarer medium (air) to a denser one (glass), it slows down and bends toward the normal; when it goes from denser to rarer, it bends away from the normal. Snell's law describes it: n1 sin(i) = n2 sin(r). Refraction explains why a straw in a glass of water seems bent, why a swimming pool looks shallower than it is, and why stars twinkle. Lenses work by refraction: a convex lens converges light and is used for magnifying glasses and to correct long-sightedness, while a concave lens diverges light and corrects short-sightedness. When light travelling in a denser medium strikes the boundary at more than the critical angle, it undergoes total internal reflection. This is the principle behind optical fibres, diamonds' sparkle and mirages.

Dispersion and Why a Rainbow Forms

Dispersion is the splitting of white light into its component colours. Isaac Newton demonstrated this in 1666 by passing sunlight through a glass prism and showing that white light is a mixture of seven colours, remembered by VIBGYOR: violet, indigo, blue, green, yellow, orange and red. Different colours bend by different amounts: violet bends the most and red the least.

A rainbow forms when sunlight is refracted, reflected and dispersed by tiny water droplets in the air. Sunlight enters a raindrop and is refracted, with each colour bending by a slightly different angle. The light is then reflected from the inside back surface of the drop by internal reflection, and refracted again as it exits. The result is that the colours are spread out, and the observer sees a circular arc. For a primary rainbow, red appears on the outside and violet on the inside, at an angle of about 42 degrees from the direction directly opposite the Sun. A rainbow is seen when the Sun is behind the observer and rain falls in front. Sometimes a fainter secondary rainbow forms above, with reversed colours. Rainbows are actually full circles, but the ground hides the lower part unless viewed from the air.

Other Optical Phenomena

Interference and diffraction demonstrate the wave nature of light. Thomas Young's double-slit experiment in 1801 produced light and dark fringes. Polarisation is used in sunglasses and 3D cinema. Scattering of sunlight by air molecules makes the sky blue, an effect named Rayleigh scattering, and sunsets appear red because blue light is scattered away over the long path of light through the atmosphere.

Optical Instruments

The human eye has a convex lens that focuses light on the retina. A camera uses lenses to focus an image on a sensor. A microscope magnifies tiny objects and a telescope brings distant objects close. Lasers produce very pure, intense, one-colour light used in surgery, barcode scanners and communication. Optical fibres carry data as pulses of light by total internal reflection, and the Indian-born physicist Narinder Singh Kapany is often called the father of fibre optics.

Light and Optics in India

Sir C. V. Raman, born in Tiruchirappalli, Tamil Nadu, discovered on 28 February 1928 that light changes its wavelength when scattered by molecules, an effect now called the Raman Effect. He won the Nobel Prize in Physics in 1930, the first Asian to win a Nobel Prize in science, and India celebrates National Science Day every 28 February. India's Thirty Meter Telescope partnership, ISRO's Aditya-L1 solar observatory and Indian Institute of Astrophysics telescopes at Kodaikanal in Tamil Nadu and Hanle in Ladakh use optics extensively.

Challenges and Future

Optics is advancing toward quantum communication, photonic computers that use light instead of electrons, and advanced telescopes like the James Webb Space Telescope.

Quick Facts

- The speed of light in vacuum is about 3 x 10^8 metres per second (299,792,458 m/s).
- Refraction is the bending of light when it passes from one medium to another because its speed changes.
- A rainbow forms by refraction, internal reflection and dispersion of sunlight in raindrops.
- C. V. Raman discovered the Raman Effect on 28 February 1928 and won the 1930 Nobel Prize.
- White light consists of seven colours, VIBGYOR.